Supermarket Sales Analysis

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Objective: Sales Performance & Customer Behavior Analysis

This analysis evaluates supermarket transaction data to identify revenue drivers, customer behavior patterns, and operational insights that support data-driven business decisions.

1. Business Context

The dataset contains 1,000 transaction-level records across three cities and branches. Each row represents a single customer purchase including product category, pricing, payment method, and rating.

The goals of this analysis are to:

- Identify key revenue drivers
- Analyze customer purchasing behavior
- Examine time-based sales patterns
- Generate actionable business recommendations

```
In [1]: import pandas as pd
        import numpy as np
        import matplotlib.pyplot as plt

        plt.style.use('default')
        pd.set_option('display.max_columns', None)
```

2. Load Dataset

```
In [2]: df = pd.read_csv("supermarket_sales.csv")
        df.head()
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	1:08:00 PM	Ewallet	522.83	4.761905	26.1415	9.1
1	226-31-3081	Giza	Yangon	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	1/8/2019	10:29:00 AM	Cash	76.40	4.761905	3.8200	9.6
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	1:23:00 PM	Credit card	324.31	4.761905	16.2155	7.4
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880	488.8480	1/27/2019	8:33:00 PM	Ewallet	465.76	4.761905	23.2880	8.4
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	10:37:00 AM	Ewallet	604.17	4.761905	30.2085	5.3

3. Dataset Overview

3.1 Dataset Dimensions

```
In [3]: df.shape
```

```
Out[3]: (1000, 17)
```

3.2 Data Types

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice ID               1000 non-null   object
 1   Branch                   1000 non-null   object
 2   City                     1000 non-null   object
 3   Customer type            1000 non-null   object
 4   Gender                   1000 non-null   object
 5   Product line             1000 non-null   object
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax 5%                   1000 non-null   float64
 9   Sales                    1000 non-null   float64
 10  Date                     1000 non-null   object
 11  Time                     1000 non-null   object
 12  Payment                  1000 non-null   object
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(8), int64(1), object(8)
memory usage: 132.8 KB
```

3.3 Summary Statistics

```
In [5]: df.describe()
```

Out[5]:

	Unit price	Quantity	Tax 5%	Sales	cogs	gross margin percentage	gross income	Rating
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1.000000e+03	1000.000000	1000.000000
mean	55.672130	5.510000	15.379369	322.966749	307.587380	4.761905e+00	15.379369	6.972700
std	26.494628	2.923431	11.708825	245.885335	234.177651	6.131498e-14	11.708825	1.71858
min	10.080000	1.000000	0.508500	10.678500	10.170000	4.761905e+00	0.508500	4.000000
25%	32.875000	3.000000	5.924875	124.422375	118.497500	4.761905e+00	5.924875	5.500000
50%	55.230000	5.000000	12.088000	253.848000	241.760000	4.761905e+00	12.088000	7.000000
75%	77.935000	8.000000	22.445250	471.350250	448.905000	4.761905e+00	22.445250	8.500000
max	99.960000	10.000000	49.650000	1042.650000	993.000000	4.761905e+00	49.650000	10.000000

3.4 Missing Values

```
In [6]: df.isnull().sum()
```

```
Out[6]: Invoice ID                 0
        Branch                     0
        City                       0
        Customer type              0
        Gender                     0
        Product line               0
        Unit price                 0
        Quantity                   0
        Tax 5%                     0
        Sales                      0
        Date                       0
        Time                       0
        Payment                    0
        cogs                       0
        gross margin percentage    0
        gross income               0
        Rating                     0
        dtype: int64
```

Observations:

- The dataset contains 1,000 transactions and 17 features.
- No missing values are present.
- Gross margin percentage is constant, indicating a fixed pricing structure.
- Revenue and gross income are proportionally related.

4. Data Cleaning & Feature Engineering

4.1 Standardize Column Names

```
In [7]: df.columns = [
            c.lower()
            .replace(" ", "_")
            .replace("%", "")
            .replace("-", "_")
            for c in df.columns
        ]

        df.columns
```

```
Out[7]: Index(['invoice_id', 'branch', 'city', 'customer_type', 'gender',
               'product_line', 'unit_price', 'quantity', 'tax_5', 'sales',
               'date', 'time', 'payment', 'cogs', 'gross_margin_percentage',
               'gross_income', 'rating'],
              dtype='object')
```

4.2 Convert Date and Create Time Features

```
In [8]: df["date"] = pd.to_datetime(df["date"])
        df["month"] = df["date"].dt.month
        df["year"] = df["date"].dt.year
        df["month_name"] = df["date"].dt.month_name()
        df["day_name"] = df["date"].dt.day_name()
        df["hour"] = pd.to_datetime(df["time"], format="%I:%M:%S").dt.hour
```

4.3 Validate Revenue Calculation

```
In [9]: difference = df["sales"] - (df["cogs"] + df["tax_5%"])
        print(difference.describe())
```

```
Out[9]: count    1.000000e+03
        mean     1.279856e-12
        std      1.136868e-13
        min      1.136868e-13
        25%      1.136868e-13
        50%      1.136868e-13
        75%      1.136868e-13
        max      4.547474e-12
        dtype: float64
```

Observation:

Minor discrepancies are due to floating-point precision and do not materially impact financial totals.

5. Key Performance Indicators (KPIs)

```
In [10]: total_revenue = df["sales"].sum()
         avg_order_value = df["sales"].mean()
         total_transactions = df["invoice_id"].nunique()
         avg_rating = df["rating"].mean()

         print("Total Revenue: {total_revenue:,.2f}")
         print("Average Order Value: {avg_order_value:,.2f}")
         print("Total Transactions: {total_transactions}")
         print("Average Rating: {avg_rating:.2f}")
```

```
Out[10]: Total Revenue: 322,966.75
         Average Order Value: 322.97
         Total Transactions: 1000
         Average Rating: 6.97
```

Insight:

The supermarket generated over 322,000 in revenue across 1,000 transactions, with an average order value of approximately 323 per transaction.

6. Revenue Analysis

6.1 Revenue by City

```
In [11]: revenue_by_city = (
             df.groupby("city")["sales"].sum().sort_values(ascending=False)
         )

         plt.figure(figsize=(8,5))
         revenue_by_city.plot(kind="bar")
         plt.title("Total Revenue by City")
         plt.ylabel("Revenue")
         plt.xticks(rotation=0)
         plt.show()
         revenue_by_city
```

```
Out[11]: City
         Naypyitaw    110568.7965
         Yangon       106197.6720
         Mandalay     106200.3120
         Name: sales, dtype: float64
```

```
In [12]: (revenue_by_city / revenue_by_city.sum) * 100).round(1)
```

```
Out[12]: City
         Naypyitaw    34.2
         Yangon       32.9
         Mandalay     32.9
         Name: sales, dtype: float64
```

Insight:

Naypyitaw generates the highest revenue, contributing approximately 34% of total sales.

6.2 Revenue by Product Line

```
In [13]: revenue_by_product = df.groupby("product_line")["sales"].sum().sort_values(ascending=False)

         plt.figure(figsize=(8,5))
         revenue_by_product.plot(kind="bar")
         plt.title("Total Revenue by Product Line")
         plt.ylabel("Revenue")
         plt.xlabel("Product Line")
         plt.xticks(rotation=45)
         plt.tight_layout()
         plt.show()

         revenue_by_product
```

```
Out[13]: product_line
         Food and beverages    56144.8800
         Sports and travel     55122.8325
         Home and lifestyle    54337.5315
         Fashion accessories   54305.8800
         Health and beauty     49193.7390
         Electronic accessories 47775.7440
         Name: sales, dtype: float64
```

Insight:

Food and Beverages generates 56.14K in revenue, the highest among all product lines, accounting for approximately 17% of total sales. Health and Beauty contributes the lowest revenue, suggesting potential underperformance relative to other categories.

6.3 Revenue Distribution

```
In [14]: plt.figure(figsize=(8,5))
         plt.hist(df["sales"],bins=40)
         plt.title("Distribution of Transaction Revenue")
         plt.ylabel("Frequency")
         plt.xlabel("Revenue")
         plt.tight_layout()
         plt.show()
```

Observation:

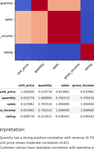
Revenue distribution is right-skewed, with most transactions clustered between 100 and 500.

7. Time-Based Sales Analysis

7.1 Daily Revenue Trend

```
In [15]: daily_sales = df.groupby("date")["sales"].sum()

         plt.figure(figsize=(10,5))
         daily_sales.plot()
         plt.title("Daily Revenue Trend")
         plt.ylabel("Revenue")
         plt.xlabel("Date")
         plt.tight_layout()
         plt.show()
```



Observation:

No consistent upward or downward trend is observed, indicating stable but non-growing short-term performance across the 3-month period.

7.2 Average Sales by Hour

```
In [16]: sales_by_hour = df.groupby("hour")["sales"].mean()

         plt.figure(figsize=(8,5))
         sales_by_hour.plot()
         plt.title("Average Revenue by Hour")
         plt.ylabel("Revenue")
         plt.xlabel("Hour")
         plt.tight_layout()
         plt.show()
```


Insight:

The highest average revenue per transaction occurs at 14:00, followed by 10:00, indicating strong late-afternoon demand. This suggests staffing and promotional focus should be optimized during peak afternoon hours.

7.3 Revenue vs Quantity

```
In [17]: plt.scatter(df["quantity"], df["sales"], alpha=0.5)
         plt.title("Revenue vs Quantity")
         plt.ylabel("Revenue")
         plt.xlabel("Quantity")
         plt.tight_layout()
         plt.show()
```


Observation:

A clear positive linear relationship is visible between quantity purchased and transaction revenue, reinforcing that revenue is primarily volume-driven.

8. Customer Behavior Analysis

```
In [18]: customer_behavior = (
             df.groupby("customer_type")["sales"]
             .agg(total_revenue="sum",
                  avg_spend="mean",
                  transactions="count")
             .round(2)
         )

         customer_behavior["total_revenue"].plot(kind="bar")
         plt.title("Total Revenue by Customer Type")
         plt.ylabel("Total Revenue")
         plt.xticks(rotation=0)
         plt.tight_layout()
         plt.show()
         customer_behavior
```



```
Out[18]:
```

customer_type	total_revenue	avg_spend	transactions
Member	169854.764	335.76245	505
Normal	153071.985	306.37297	495

Insight:

Member customers generate higher total revenue and higher average spending per transaction, demonstrating the effectiveness of the membership program.

9. Correlation Analysis

```
In [19]: corr = df[["unit_price","quantity","sales","gross_income","rating"]].corr()

         plt.figure(figsize=(6,5))
         plt.imshow(corr, cmap="coolwarm")
         plt.colorbar()
         plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
         plt.yticks(range(len(corr.columns)), corr.columns)
         plt.tight_layout()
         plt.show()
         corr
```



```
Out[19]:
```

	unit_price	quantity	sales	gross_income	rating
unit_price	1.000000	0.010778	0.633962	0.633962	-0.008778
quantity	0.010778	1.000000	0.705510	0.705510	-0.015815
sales	0.633962	0.705510	1.000000	1.000000	-0.036442
gross_income	0.633962	0.705510	1.000000	1.000000	-0.036442
rating	-0.008778	-0.015815	-0.036442	-0.036442	1.000000

Interpretation:

- Quantity has a strong positive correlation with revenue (0.70).
- Unit price shows moderate correlation (0.63).
- Customer rating shows no meaningful relationship with operating behavior.
- Gross income scales proportionally with revenue due to fixed margins.

10. Executive Summary

Revenue performance is primarily volume-driven rather than price-driven. Member customers contribute higher transaction value, reinforcing the strategic importance of loyalty programs. Peak revenue occurs during afternoon and early evening hours, supporting optimized staffing and promotional scheduling.

11. Business Recommendations

- Prioritize marketing investment in high-performing product lines.
- Expand membership program to increase customer lifetime value.
- Optimize staffing during peak afternoon hours.
- Implement dashboard monitoring for continuous performance tracking.

12. Limitations

- Dataset covers only three months, limiting seasonal analysis.
- No marketing campaign or discount data included
- Analysis is descriptive and does not include predictive modeling